Ericsson PE Node (BAS-PE-10) Optical Transport Failure (ALARM_8893) Resolution SOP

Node: BAS-PE-10 | **Vendor:** Ericsson | **Severity:** Critical | **Alarm Code:** ALARM_8893

Description

This Standard Operating Procedure (SOP) outlines the steps to be taken to troubleshoot and resolve the Optical Transport Failure (ALARM_8893) on the Ericsson PE Node (BAS-PE-10) at the Basavanagudi POP. The failure domain is primarily caused by SFP/XFP degradation, fiber cuts, or Rx/Tx power loss. The PE Node is responsible for terminating the optical transport network, and the failure of this node can disrupt the entire network. The underlying protocol mechanics involve the Optical Transport Network (OTN) protocol, which is used for transmitting data over optical fibers. The OTN protocol uses SFP/XFP transceivers to convert electrical signals to optical signals and vice versa. The failure of these transceivers or the fiber itself can cause the ALARM_8893 to be raised. The root cause of the failure can be a hardware issue, such as a faulty SFP/XFP, or a software issue, such as misconfigured OTN parameters.

Pre-requisite Checks

- Check: Verify that the Basavanagudi POP is accessible and the remote hands are dispatched.
- Check: Check the redundant link status to ensure that it is operational.
- Check: Verify that the OTN parameters are correctly configured.

Potential Root Causes

Cause	Probability
Faulty SFP/XFP	High
Fiber Cut	Medium
Misconfigured OTN Parameters	Low

Diagnostic Commands & Expected Outputs

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Command: show opt tx, Expected Output: Optical transmit parameters, Description: This command displays the optical transmit parameters, including the SFP/XFP type, transmit power, and wavelength. It helps to identify any issues with the SFP/XFP or the fiber.

Command: show opt rx, Expected Output: Optical receive parameters, Description: This command displays the optical receive parameters, including the SFP/XFP type, receive power, and wavelength. It helps to identify any issues with the SFP/XFP or the fiber.

Command: show opt link, Expected Output: Optical link parameters, Description: This command displays the optical link parameters, including the link status, transmit power, and receive power. It helps
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to identify any issues with the fiber or the SFP/XFP.
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Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP, Command: replace opt tx, Description: Replace the faulty SFP/XFP with a new one. Ensure that the new SFP/XFP is compatible with the PE Node and the OTN protocol.

Step 2: Step: Reconfigure the OTN parameters, Command: config opt param, Description: Reconfigure the OTN parameters to ensure that they are correctly set. This includes setting the transmit power, receive power, and wavelength.

Step 3: Step: Verify the fiber connectivity, Command: verify opt link, Description: Verify the fiber connectivity by checking the link status, transmit power, and receive power. Ensure that the fiber is not damaged and the SFP/XFP is securely connected.

Post-Check Verification

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Command: show opt tx, Expected Output: Optical transmit parameters (correct), Description: Verify that the optical transmit parameters are correctly set after replacing the SFP/XFP or reconfiguring the OTN parameters.

Command: show opt rx, Expected Output: Optical receive parameters (correct), Description: Verify that the optical receive parameters are correctly set after replacing the SFP/XFP or reconfiguring the OTN parameters.

Command: show opt link, Expected Output: Optical link parameters (correct), Description: Verify that the optical link parameters are correctly set after verifying the fiber connectivity.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP, Command: restore opt tx, Description: Restore the original SFP/XFP if the replacement fails.

- Step: Revert the OTN parameters, Command: revert opt param, Description: Revert the OTN parameters to their original settings if the reconfiguration fails.

- Step: Verify the original fiber connectivity, Command: verify opt link, Description: Verify the original fiber connectivity by checking the link status, transmit power, and receive power.